Spatial–Frequency Dual-Domain Feature Fusion Network for Low-Light Remote Sensing Image Enhancement

Zishu Yao, Guodong Fan, Jinfu Fan, Min Gan, *Senior Member, IEEE*, and C. L. Philip Chen, *Fellow, IEEE*

***Abstract*— Low-light remote sensing (RS) images generally feature high resolution and high spatial complexity, with continuously distributed surface features in space. This continuity in scenes leads to extensive long-range correlations in spatial domains within RS images. convolutional neural networks (CNNs), which rely on local correlations for long-distance modeling, struggle to establish long-range correlations in such images. On the other hand, transformer-based methods that focus on global information face high computational complexities when processing high-resolution RS images. From another perspective, the Fourier transform can compute global information without introducing a large number of parameters, enabling the network to more efficiently capture the overall image structure and establish long-range correlations. Therefore, we propose a dual-domain feature fusion network (DFFN) for low-light RS image enhancement. Specifically, this challenging task of low-light enhancement is divided into two more manageable subtasks: the first phase learns amplitude information to restore image brightness, and the second phase learns phase information to refine details. To facilitate information exchange between the two phases, we designed an information fusion affine block that combines data from different phases and scales. In addition, we have constructed two dark light RS datasets to address the current lack of datasets in dark light RS image enhancement. Extensive evaluations show that our method outperforms existing state-of-the-art methods. The code is available at https://github.com/iijjlk/DFFN.**

***Index Terms*— Computer vision, frequency, image enhancement, remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) technology aims to capture clear ground scene information. Compared to ordinary images, RS images display richer color information and broader spatial information. However, due to nighttime conditions and other inherent natural factors, images captured in low-light environments often suffer from color degradation and loss of detail. This inevitably affects many downstream tasks [1], [2] that rely on RS images, such as disaster recovery, wildlife monitoring, and environmental protection. Therefore, the development of algorithms dedicated to enhancing low-light RS images is crucial.

Compared to images in traditional low-light image enhancement tasks [3], [4], [5], [6], [7], [8], [9], RS images have larger lengths and widths. Some surface features are often continuously distributed in space, such as forests covering several square kilometers or roads traversing the entire image. The continuity of such scenes results in RS images exhibiting long-distance correlations in spatial aggregation. Convolutional neural networks (CNNs) [10], [11], [12] rely on local perception. For high-resolution images, the receptive field of a single convolutional operation struggles to capture larger spatial structures or global information in the image. Although deep convolutions can expand the receptive field, as the network depth increases, the loss of detailed information gradually occurs, making it challenging for models to establish long correlations in RS images. Recently, a number of transformer methods [17], [18], [19], [20], [21], with vision transformer [22] as a prominent example, have successfully addressed the challenge of establishing long-range dependencies by implementation of self-attention mechanisms. However, the quadratic complexity problem remains difficult to solve. In addition, these methods tend to overfit when trained on small datasets, significantly reducing their reliability in practical applications. Fig. 1(a) presents visual results and a number of parameters for several representative methods. It can be observed that SNR, built upon the transformer, achieves suboptimal visual results, yet concurrently bears an onerous computational burden.

Most existing methods [23], [24], [25] process images in the spatial domain based on pixel perspectives. For instance, Xing et al. [26] proposed an unsupervised GAN framework for low-light RS image enhancement based on the self-similarity characteristics of RS images. Singh et al. [27] designed a multiresolution Rnet that preserves high-resolution information while retaining low-resolution standards through different data streams. In contrast, we shift our focus to the Fourier domain, motivated by two unique phenomena inherent in the Fourier domain: 1) under Fourier transformation, global information of the image can be extracted without introducing a large number of parameters [28]. 2) It is well-known that the Fourier phase

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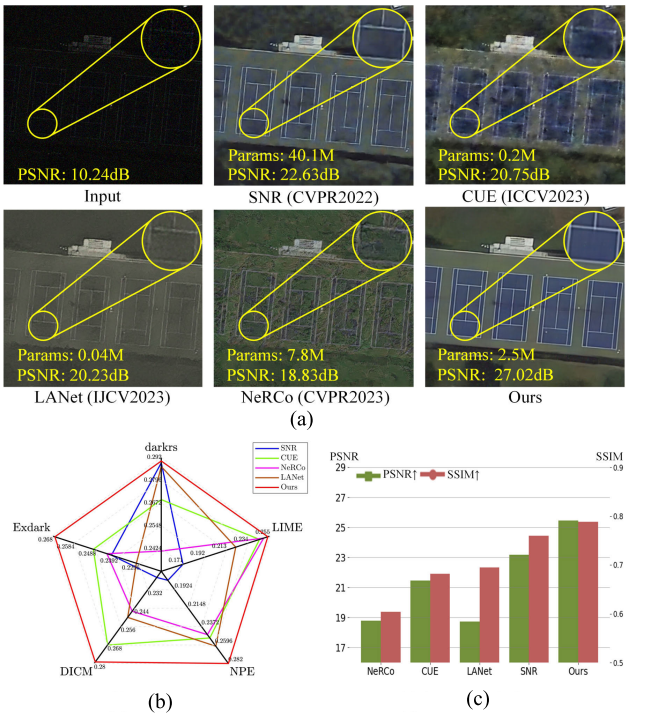


Fig. 1. Comparison between the latest state-of-the-art methods and our approach. As shown in the zoomed-in view region, these methods exhibit color distortion and excessive noise, leading to reduced visual quality. In contrast, our method presents vibrant colors and sharp outlines. Transformer-based method SNR [13] achieves similar visual effects to ours but with 40.1M parameters. In addition, in (b), we compute the NIQE [14] scores on a no-reference dataset and display them in inverse form. Furthermore, in (c), scores of SSIM [15] and PSNR [16] metrics on a reference dataset are shown. It can be easily observed that our method significantly outperforms others. (a) Visual quality. (b) NIQE score. (c) iSAID-dark dataset.

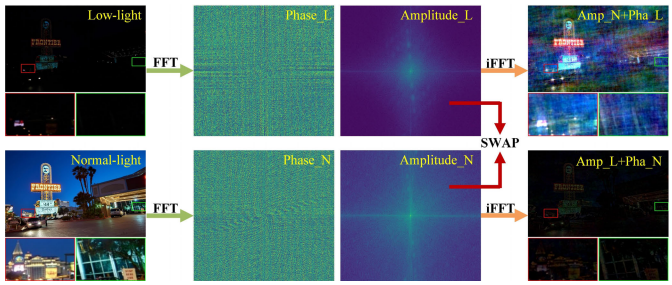


Fig. 2. Our motivation. As observed in the figure, when swapping the amplitude and phase of low-light images with those of normal images, the overall ambient light of the image containing degraded amplitude darkens while maintaining clear details. On the other hand, in the image containing phase degradation, there is no change in brightness, but the image details vanish. This indicates that the Fourier transform can decouple degradation information.

preserves high-level semantics, while amplitude contains low-level features [29]. Fig. 2 displays the visualization image of exchanging the Fourier amplitude and phase between normal and low-light images. It can be observed that brightness degradation exists in the amplitude, while other information is preserved in the phase. This phenomenon suggests that the Fourier transform can decouple the degradation of low-light images, breaking down the complex coupling problem into two relatively easier-to-solve subproblems. Existing methods have explored the frequency domain information, but still present many shortcomings. For example, Wang et al. [28] opt to simultaneously recover both amplitude and phase information without considering spatial information. Li et al. [30] only perform Fourier decomposition on images within the network, yet the loss function continues to make comparisons with the ground truth (GT) solely in the spatial domain, lacking effective supervision. This undoubtedly makes the recovery process more difficult. Therefore, the challenge of how to integrate frequency domain information with spatial features and efficiently apply this integration for the enhancement of RS low-light images remains unresolved.

In addition, multistage architecture networks [4], [28], [31], [32] often face issues with insufficient information interaction. Most existing works directly use the output of the previous stage as the input for the next stage. However, such methods do not consider the rich information contained in the intermediate features, which can easily lead to information loss. Other works adopt a one-to-one correspondence approach to pass information from the previous stage to the next stage, effectively compensating for the information loss and detail blurring caused by the upsampling and downsampling processes. However, these solutions still only consider the interaction of information at the same scale, neglecting the interaction of cross-scale features within and between stages, making it difficult to obtain more refined contextual features.

In this work, based on two unique phenomena of the Fourier transform, we propose a dual-domain feature fusion network (DFFN). Specifically, the DFFN consists of two-stage transformations to progressively restore high-quality images. Initially, the amplitude illumination phase learns the amplitude mapping from low-light to normal lighting conditions to brighten the image. Subsequently, the phase refinement phase enhances image details further. To integrate the frequency domain and spatial features, we designed the dual-domain amplitude block (DDAB) for the amplitude illumination phase and the dual-domain phase block (DDPB) for the phase refinement phase. By splitting the Low-light Remote Image Enhancement task into two subproblems, we reduce the complexity of degradation coupling.

Meanwhile, to improve the information interaction capability between the two-stage networks, we designed an information fusion affine module (IFAM) that integrates rich feature information from different stages and scales, achieving adaptive learning through dynamic fusion with affine filtering. The obtained feature information is convolved with the decoder features of the phase refinement stage to enhance the network representation capability.

In addition, due to the lack of available image pairs, the development of existing RS low-light enhancement tasks has been slow. In this article, we constructed two large datasets, with the paired dataset iSAID-dark used for training and the real dataset darkrs used for testing.

In summary, our work contributes the following.

1) We propose a DFFN for low-light RS image enhancement, which decomposes the degradation problem into two relatively easier-to-solve subproblems through a two-stage network.

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2) We design an IFAM to facilitate information interaction across scales, stages, and domains within the two-stage network. This module promotes information fusion between the two stages and enhances the representation of the global context.
3) We construct two low-light RS datasets, including a real dataset captured by UAVs at night and a large-scale synthetic dataset. These datasets provide paired data for low-light RS image enhancement and real low-light scenes for testing purposes.
4) We conduct extensive experiments on multiple public benchmarks for a comprehensive evaluation, demonstrating that our method performs well compared to state-of-the-art methods. In addition, our approach shows excellent tradeoffs between model complexity and performance.

II. Related Work

A. Low-Light Image Enhancement

In recent years, low-light image enhancement has developed rapidly, which can mainly be classified into three categories: methods based on value mapping, model-based methods, and data-driven methods.

Value mapping-based methods enhance image exposure by altering the pixel distribution through mapping relationships. Notably, such methods include histogram equalization [33], [34], [35] and curve transformations, such as gamma transformation [36], [37]. However, due to the lack of consideration for pixel context information, these methods are prone to causing local illumination inconsistencies and color abnormalities.

Model-based methods [38], [39], [40], grounded in Retinex theory, view images as the product of reflectance and irradiance. By dividing low-light images by the irradiance image, the resulting reflectance is used as the final enhanced image. For example, Li et al. [41], building upon traditional Retinex theory and taking noise maps into account, successfully developed a noise-robust Retinex model. Guo et al. [42] estimated the illumination for each pixel using the maximum channel prior and added a prior structure on top of it to refine the illumination. The crux of such methods lies in designing well-crafted regularization functions. However, these regularization functions are based on prior assumptions, which, when not valid, greatly diminish the effectiveness of these methods.

Data-driven approaches [43], [44], [45] are based on large-scale datasets and enhance network performance through the design of various complex model structures. For instance, Ma et al. [8] proposed a context-sensitive decomposition network that focuses on the contextual dependencies in the feature space, estimating lighting and reflections at the feature level to ultimately achieve high-quality enhancement effects. Li et al. [46] designed a pixel-level curve estimation method and developed a series of loss functions that, while ensuring high-speed computation, yielded excellent results. Wang et al. [47] combined gamma correction with transformers, establishing dependencies for all pixels from local to global, efficiently recovering dark areas.

Although existing low-light image enhancement methods have achieved remarkable enhancement effects, due to the lack of corresponding paired datasets and the rarity of methods specifically designed for RS images, current mainstream methods struggle to achieve ideal results in the task of enhancing dark light RS images. Recently, Xing et al. [26] proposed an unpaired training method for low-light RS image enhancement, which maximizes the mutual information between dark and normal light images in a generative adversarial network through self-similarity-based contrastive learning. However, due to limited supervision, the final enhancement effect is not satisfactory. To facilitate the development of this task, we propose a large-scale paired dataset for dark light RS image enhancement to provide effective supervision.

B. Multistage Network

The multistage network is designed to decompose challenging tasks into several easier subtasks and solve them progressively to achieve optimal results. However, current multistage networks still have some issues. Specifically, existing multistage networks can be divided into the following two categories: 1) result fusion and 2) cross-level fusion.

The first category often uses the output of the previous stage as the input for the next stage. For example, Zhu et al. [31] proposed an edge-enhanced multiexposure fusion network (EEMEFN), where they initially brighten dark images and then enhance the edges of the preliminary enhanced results to obtain the final restored outcome. Li et al. [4] utilized a recurrent network to learn low-light image enhancement, with each stage taking the attention map and the previous stage's output as input, and then performing recursive learning through the same residual module. Wang et al. [28], in their proposed dual-stage network, first enhanced image brightness by estimating magnitude information, followed by the introduction of a signal-to-noise ratio (SNR) map to restore image details. Although these methods achieve good visual effects, they all neglect the rich intermediate features and solely using the preliminarily enhanced image can lead to information loss, resulting in blurred details.

The second category aims to integrate rich contextual features across multiple stages, alleviating information loss and gradient vanishing issues in deep networks. For instance, in the proposed deep generalized unfolding network (DGUNet), Mou et al. [32] passed the decoder features from each stage to the corresponding module in the next stage to capture rich informational features. However, these methods typically only consider the interactions between paired features and do not account for feature interactions across different scales.

In contrast, our proposed IFAM effectively combines feature information across different stages and scales. By utilizing dynamic fusion and affine filtering for adaptive learning, it significantly enhances the network's representational capacity.

C. Fourier Transform in Neural Networks

Recently, due to the unique properties of images in the Fourier space, such as capturing global representations, Fourier frequency information [28], [30], [48] has attracted widespread attention. A series of works have been based

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on Fourier domain transformations to enhance the generalization performance of networks. For instance, Mao et al. [49] designed a residual Fourier block to address issues like the potential neglect of low-frequency information and the difficulty in modeling long-distance information in Resblocks. Suvorov et al. [50] used fast Fourier convolution to design a large mask inpainting network, achieving impressive results even in challenging scenarios. Fuoli et al. [51] employed Fourier loss to aid networks in recovering high-frequency information in super-resolution images. Meanwhile, the field of RS has seen extensive exploration of frequency domain information [52]. For example, Zhou et al. [53] demonstrated excellent performance in the pan-sharpening task by exploring the relationship between Pan images and low-resolution multispectral (MS) images with frequency domain information. Yu et al. [54] proposed a spiking neural network in the frequency domain (SNNFD), which enhances the model's transferability and feature extraction capability for buildings of different sizes by integrating frequency and spatial domain learning. Inspired by these works, we proposed DFFN based on the Fourier transform, which effectively decouples low-light degradation and achieves superior enhancement results.

III. Method

In this section, we first introduced the Fourier information transformation. Subsequently, we described the overall architecture of the DFFN and the basic units constituting the network. Finally, we elaborated on the IFAM and the loss functions used in the DFFN.

A. Fourier Frequency Information

In this section, we will briefly introduce Fourier frequency information. Specifically, consider an image $I \in \mathbb{R}^{H\times W\times 1}$, whose Fourier transformation function $\mathcal{F}$ can be represented as

$$\mathcal{F}(X)(i,j) = \frac{1}{\sqrt{HW}} \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} X(h,w) e^{-j2\pi\left(\frac{h}{H}i + \frac{w}{W}j\right)} \tag{1}$$

where k represents the imaginary unit, i and j represent the horizontal and vertical coordinates, respectively, h and w represent spatial coordinates, and the inverse process of $\mathcal{F}$ is denoted as $\mathcal{F}^{-1}$.

Correspondingly, the amplitude $\mathcal{A}(I)$ and phase $\mathcal{P}(I)$ information can be defined as

$$\begin{cases} \mathcal{A}(I)(i,j) = \sqrt{R^2(X)(i,j) + I^2(X)(i,j)} \\ \mathcal{P}(I)(i,j) = \arctan\left[\dfrac{I(X)(i,j)}{R(X)(i,j)}\right] \end{cases} \tag{2}$$

where $I(X)(i,j)$ and $R(X)(i,j)$ denote the imaginary and real component of $F(X)(i,j)$. For feature maps or color images, Fourier calculations are computed separately for each channel. The $R(X)(i,j)$ and $I(X)(i,j)$ can also be obtained by

$$\begin{cases} R(X)(i,j) = \mathcal{A}(X)(i,j) \times \cos(\mathcal{P}(X)(i,j)) \\ I(X)(i,j) = \mathcal{A}(X)(i,j) \times \sin(\mathcal{P}(X)(i,j)). \end{cases} \tag{3}$$

B. Network Architecture

The structure of DFFN is shown in Fig. 3. Specifically, in the amplitude illumination stage, image brightness is restored by adjusting the amplitude, while in the phase refinement stage, phase information is learned to refine fine details. Since the two stages have different subtasks, their input information and supervision targets should also differ. The low-light image I_{low} as the input for the first stage. The supervision target is defined as $\mathcal{F}^{-1}(\mathcal{A}(I_{\text{gt}}), \mathcal{P}(I_{\text{low}}))$. To prevent the first stage network from affecting the phase information, the output O_A is not directly used as the input for the second stage. Instead, let $I_{\text{mix}} = \mathcal{F}^{-1}(\mathcal{A}(O_A), \mathcal{P}(I_{\text{low}}))$, and the supervision target is I_{gt}.

Although our primary motivation lies within the Fourier domain, the utilization of spatial branches is necessary. This is because spatial branches and Fourier branches complement each other. Spatial branches, employing convolution operations, effectively model structural dependencies in the spatial domain. Fourier branches can engage in global information, facilitating the unraveling of energy and degradation. Therefore, for more efficient information recovery, we introduce DDAB and DDPB.

1) Dual-Domain Amplitude Block: As shown in the bottom-left of Fig. 3, the DDAB consists of two parallel branches: the spatial domain branch and the frequency domain branch. The spatial domain branch has two 3×3 Conv layers to capture spatial information. In the frequency domain branch, the feature stream F_{in} first undergoes 1×1 Conv to refine the features. Then, FFT is applied to obtain the amplitude component $\mathcal{A}(F_{\text{in}})$ and the phase component $\mathcal{P}(F_{\text{in}})$. The amplitude component $\mathcal{A}(F_{\text{in}})$ is passed through a Conv layer with two 1×1 kernels (to avoid damaging the frequency domain structure, only 1×1 Conv is used here). Meanwhile, the corresponding phase information $\mathcal{P}(F_{\text{in}})$ is preserved through a separate branch. Subsequently, we use the iFFT to map $\mathcal{A}(F_{\text{in}})$ and $\mathcal{P}(F_{\text{in}})$ back to the image space to obtain frequency domain features. Finally, we integrate the spatial and frequency domain features through residual connections. The process can be described as follows:

$$\begin{cases} F_{\text{sa}} = \sigma_{3\times 3}\left[\sigma_{3\times 3}(F_{\text{in}})\right] \\ \mathcal{A}(F_{\text{in}}), \mathcal{P}(F_{\text{in}}) = \mathcal{F}[\Theta_{1\times 1}(F_{\text{in}})] \\ F'_A = \mathcal{F}^{-1}\langle \Theta_{1\times 1}\{\sigma_{1\times 1}[\mathcal{A}(F_{\text{in}})]\}, \mathcal{P}(F_{\text{in}})\rangle \\ F_A = F_{\text{sa}} + F'_A \end{cases} \tag{4}$$

where $\sigma(\cdot)$ represents Conv and ReLU operation, $\mho(\cdot)$ represents concatenation operation, and $\Theta(\cdot)$ represents Conv operation.

2) Dual-Domain Phase Block: Compared to DDAB, DDPB simply swaps the operations of amplitude and phase. It can be described as

$$F'_P = \mathcal{F}^{-1}\langle \Theta_{1\times 1}\{\sigma_{1\times 1}[\mathcal{P}(F_{\text{in}})]\}, \mathcal{A}(F_{\text{in}})\rangle. \tag{5}$$

In summary, the amplitude illumination stage utilizes DDAB as the basic unit and employs an encoder–decoder structure to extract multiscale feature information. Similarly, the phase refinement stage adopts DDPB as the basic unit. Skip connections are established between the corresponding encoder

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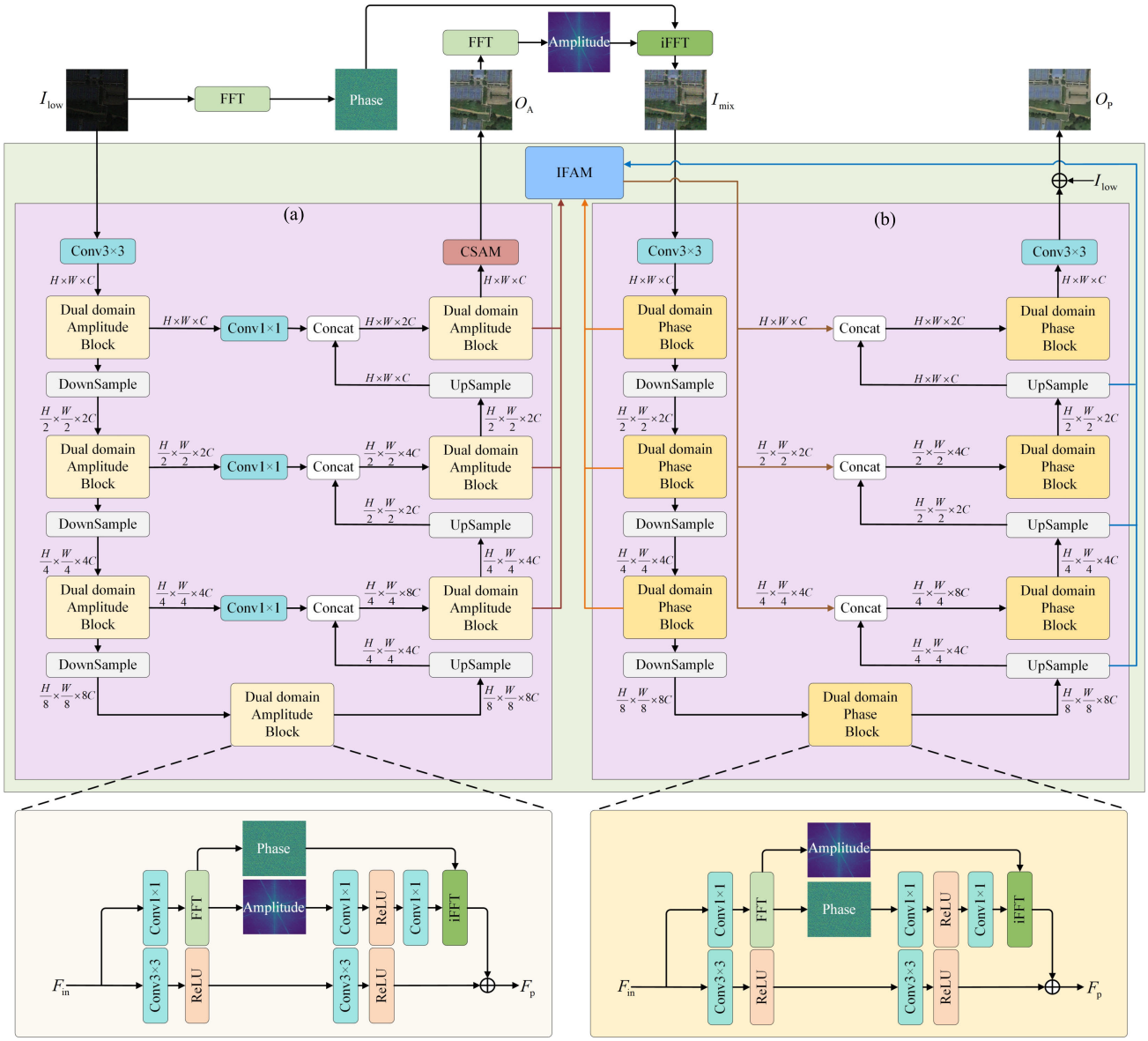


Fig. 3. Overall network structure diagram. The specific structures of DDAB and DDPB are displayed below the diagram. The low-light image I_{low} first undergoes an amplitude illumination stage to obtain a preliminary enhanced image O_A, followed by a phase refinement stage to obtain the final enhanced image O_P. In this process, the feature information from different stages is passed into IFAM for mapping to enhance the contextual representation capability of the model. (a) Stage 1: amplitude illumination. (b) Stage 2: phase refinement.

and decoder at each scale of the amplitude illumination stage. In contrast, the phase refinement stage employs IFAM to map features from the previous stage and the current encoder features more efficiently for fusing information between the two stages. Furthermore, to optimize the cross-stage network, a cross-stage feature fusion, and supervised attention module (CSAM) [55] is introduced at the end of the amplitude illumination stage.

Next, we will provide a detailed explanation of the proposed IFAM module and loss function.

C. Information Fusion Affine Module

Multistage networks [4], [28], [51] enhance performance by decomposing complex problems into simpler subproblems and solving them separately. In recent years, various works have achieved significant results using this approach. However, improving the ability of cross-stage information interaction remains a major challenge for multistage networks. As the depth of the network increases, shallow features tend to be lost gradually. Simply passing the output from the previous stage to the next stage may result in the loss of a significant amount of potential features. Another mainstream approach is to transfer information between stages of the same size, but this lacks interaction between features of different scales.

Therefore, our designed IFAM first aggregates feature information from different stages and scales. Subsequently, it performs adaptive fusion on the aggregated features to enhance global contextual representation. Specifically, IFAM (as shown in Fig. 4) comprises two submodules.

1) Information Fusion Module: IAM integrates all scale information from the first-stage decoder and second-stage encoder, and merges information from different scales.

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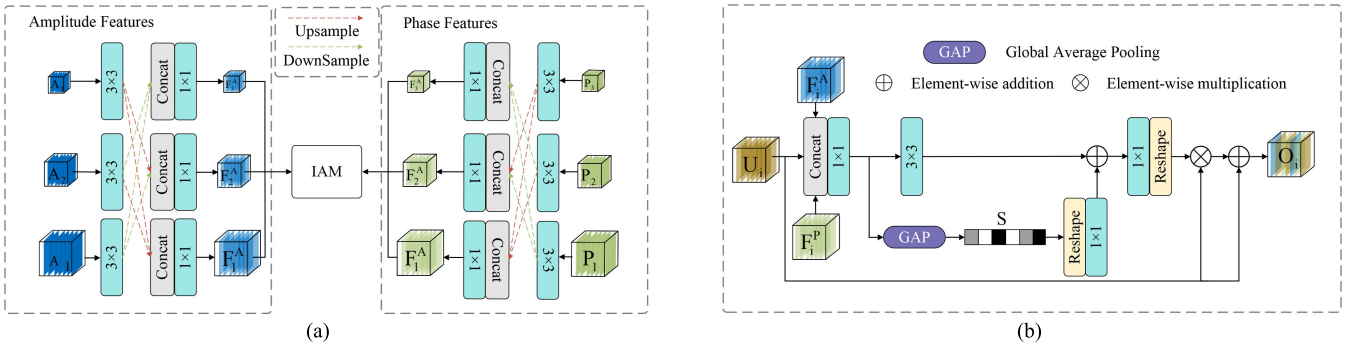


Fig. 4. Specific structure of the proposed IFAM consists of two key components: the information fusion module and the information mapping module. U_i represents the feature obtained after the ith upsampling in the phase refinement stage. (a) Information fusion module. (b) Information affine module.

As shown in Fig. 4(a), the decoder features $\{A_i\}_{i=1}^{3}$ first undergo a 3×3 Conv, followed by combining different scale features through upsample and downsample operations. Then, 1×1 Conv is used to merge these features with different receptive fields, obtaining rich contextual information. The encoder features $\{P_i\}_{i=1}^{3}$ undergo similar operations. Subsequently, the fused features are passed into IAM.

2) Information Affine Module: As shown in Fig. 4(b), taking the feature information from the ith layer as an example, three features of the same size from different encoders and decoders are first fused through a 1×1 Conv. Subsequently, they pass through separate 3×3 Conv to obtain spatial context information, and through a pooling layer and 1×1 Conv to obtain channel context information. Finally, the two streams of information are added element-wise to fully utilize their information

$$\begin{cases} F_{\text{fuse}} = \Theta[\mho(A_i, P_i, U_i)] \\ F'_{\text{fuse}} = \Theta(F_{\text{fuse}}) + \text{Pool}\{\text{Reshape}[\Theta(F_{\text{fuse}})]\} \end{cases} \tag{6}$$

where U_i represents the decoder features from the second stage.

Subsequently, weight filters $F_{\text{sv}} \in \mathbb{R}^{(k\times k\times c)\times h\times w}$ are generated based on 1×1 Conv, and reshaped into independent pixel kernels $F'_{\text{sv}} = \{W_{i,j} \mid i \in [1, h], j \in [1, w]\}$, where $W_{i,j} \in \mathbb{R}^{k^2\times c}$. In this way, each position is adjusted to the optimum with adaptive weights. The filtering result can be represented as

$$O_i = U_i \otimes F'_{\text{sv}} + U_i \tag{7}$$

where $\otimes$ indicates the element-wise multiplication.

Through the above process, IFAM generates adaptive weight filtering guided by multistage, multiscale, and cross-domain interactive features, enabling the generated features to adapt more flexibly to image content.

D. Loss Function

Due to our DFFN having two outputs O_A and O_P, the loss functions L_A and L_P for the two stages are as follows:

$$\begin{cases} L_A = \left\| O_A - \mathcal{F}^{-1}(\mathcal{A}(I_{\text{gt}}), \mathcal{P}(I_{\text{low}}) \right\|_1 + \alpha \left\| \mathcal{A}(O_A) - \mathcal{A}(I_{\text{gt}}) \right\|_1 \\ L_P = \left\| O_P - I_{\text{gt}} \right\|_1 + \beta \left\| \mathcal{F}(O_P) - \mathcal{F}(I_{\text{gt}}) \right\|_1 \\ \qquad + \gamma \left\| \mathcal{P}(O_P) - \mathcal{P}(I_{\text{gt}}) \right\|_1 \end{cases} \tag{8}$$

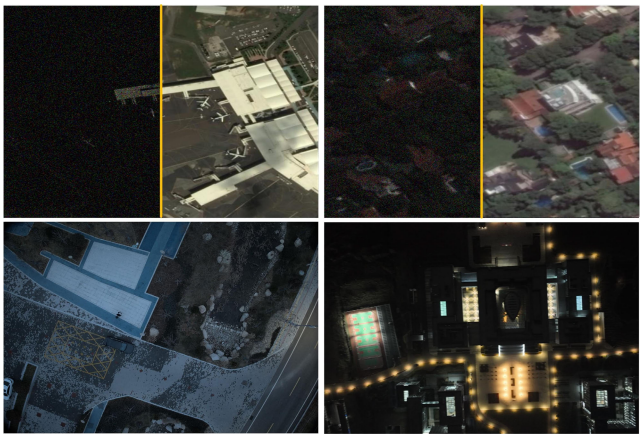


Fig. 5. Samples from the proposed iSAID-dark (top) and darkrs (bottom) dataset.

where $\|\cdot\|_1$ presents the mean absolute error (MAE), α, β, and γ are the weight factor, and we empirically set them to 0.05. Therefore, the overall loss is

$$L = L_A + L_P. \tag{9}$$

IV. Dataset

Due to the particularity of RS images, collecting paired normal/low-light image sets is almost an impossible task. In this work, inspired by the significant achievements of previous synthetic datasets in various low-quality image enhancement tasks, we introduced a synthesis mechanism to generate the corresponding low-light RS dataset. In addition, to validate the reliability of the proposed DFFN and iSAID-dark in real low-light environments, we collected real nighttime RS images by capturing different scenes with drones at night, forming a test dataset named dark RS (darkrs), which comprises 86 test images. The two proposed datasets are shown in Fig. 5.

A. Data Collection

The iSAID [61] dataset has collected a large number of high-resolution aerial images from various scenes including rural and urban areas. Therefore, we selected 751 images from the iSAID test set as our benchmark data. Considering the significant differences in size and shape of RS images, and

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TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE REFERENCE DATASET, WITH THE BEST RESULTS INDICATED IN **BOLD** AND THE SECOND-BEST RESULTS UNDERLINED

Methods	Venue	LOL			iSAID-dark			iSAID-dark (retrain)			iSAID-dark (high-pixel)		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Zero-DCE++ [46]	TPAMI'22	12.17	0.511	0.258	12.07	0.040	0.818	8.46	0.055	0.950	12.20	0.157	0.920
SCI [56]	CVPR'22	14.78	0.525	0.238	12.14	0.194	0.901	13.08	0.205	0.745	13.73	0.168	0.854
SNR [13]	CVPR'22	21.22	**0.834**	0.097	13.31	0.246	0.722	23.17	0.760	0.179	22.06	0.711	0.238
LLFormer [57]	AAAI'23	23.25	0.819	0.105	16.66	0.376	0.658	23.11	0.725	0.221	-	-	-
CUE [58]	ICCV'23	21.68	0.769	0.151	15.70	0.343	0.717	21.47	0.682	0.263	19.16	0.631	0.349
FourLLIE [28]	ACMM'23	20.02	0.818	0.088	14.49	0.350	0.579	19.78	0.643	0.309	17.58	0.503	0.302
LANet [59]	IJCV'23	21.74	0.816	0.101	11.88	0.089	0.804	18.72	0.695	0.231	21.95	0.690	0.247
NeRCo [60]	CVPR'23	22.94	0.783	0.102	**16.73**	**0.513**	**0.381**	18.79	0.604	0.326	20.94	0.568	0.377
Ours	-	**23.46**	**0.834**	**0.087**	15.51	0.287	0.714	**25.30**	**0.784**	**0.151**	**22.47**	**0.721**	**0.192**

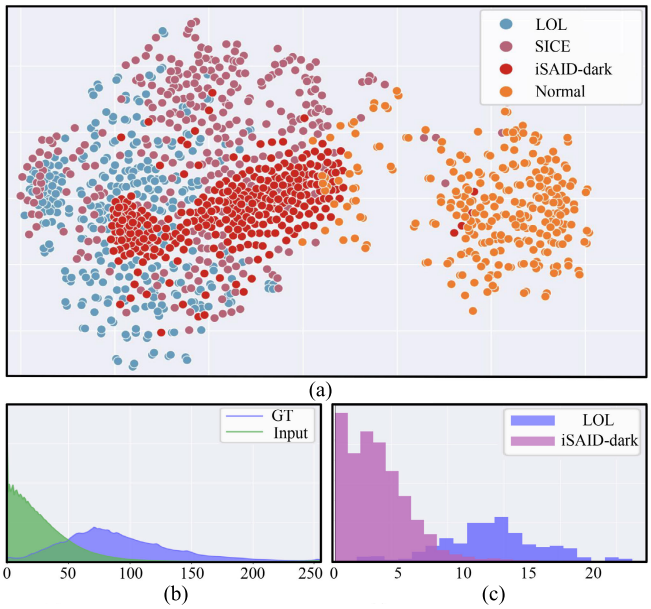


Fig. 6. Distribution analysis of the iSAID-dark dataset. (a) T-SNE map indicates that the proposed dataset exhibits a broader data range. (b) Brightness range of normal-light (blue) and low-light images (green). (c) Range distribution of SNR values for both the LOL dataset and the proposed dataset is shown. It can be observed that the proposed dataset has a lower SNR range, indicating that its noise levels are more challenging.

the difficulty of training high-resolution images in mainstream networks, and in order to increase the diversity of scenes included in the dataset, we performed multiple random crops on these 751 images. The cropped images are resized to 500 × 500. Subsequently, we divided them into 3755 training image pairs and 66 validation image pairs. To further validate the recovery performance at high resolutions, we constructed a corresponding high-resolution dataset called iSAID-dark (high-pixel), which includes 72 test images, all with a resolution of 1080p.

B. Data Generation

1) Low-Light Simulation: Inspired by Zero-DCE++ [46], which treats enhancing illumination as an estimation task of specific curves for images and achieves excellent results, we adopted an inverse transformation of Zero-DCE++ to simulate low-light images by adjusting the reverse curve. This allows us to generate low-light images with adjustable darkness levels. Specifically, we set a random darkness range to replace the fixed exposure value of brightness loss in Zero-DCE++. By using different exposure levels, the trained reverse Zero-DCE++ can generate low-light images with varying darkness levels.

2) Noise Simulation: To simulate more realistic low-light images, we also added Gaussian noise of varying degrees to the generated images, all of which followed a normal distribution. In addition, for images with higher levels of darkness, the added noise was more intense.

C. Data Distribution

Fig. 6(a) displays the T-SNE visualization results of our proposed iSAID-dark dataset compared to other datasets, demonstrating that iSAID-dark encompasses common features of LOL and SICE, indicating a broader data range in our dataset. Fig. 6(b) illustrates the brightness distribution of our proposed dataset. Meanwhile, we plotted the SNR distribution of the proposed dataset and the LOL dataset, as shown in Fig. 6(c), to demonstrate the noise levels in the images. The SNR distribution indicates that our dataset possesses a wide and challenging SNR range.

D. Discussion

Our dataset simulates real low-light environments, encompassing the majority of common RS scenes. In addition, our randomly set dark intensity range and noise cover the most common challenging low-light scenarios. When given different exposure strengths, the reverse Zero-DCE++ can generate low-light images with different brightness levels. Fig. 7 illustrates images with different levels of low-light and noise.

V. EXPERIMENTS

In this section, we first describe the detailed setup of the experiments, including the datasets used, evaluation metrics, comparison methods, and implementation details. Subsequently, we quantitatively and qualitatively evaluate the proposed DFFN against state-of-the-art methods on the proposed low-light RS dataset and real-world low-light dataset. In addition, we provide extensive ablation studies to validate the effectiveness of the proposed DFFN architecture and its components.

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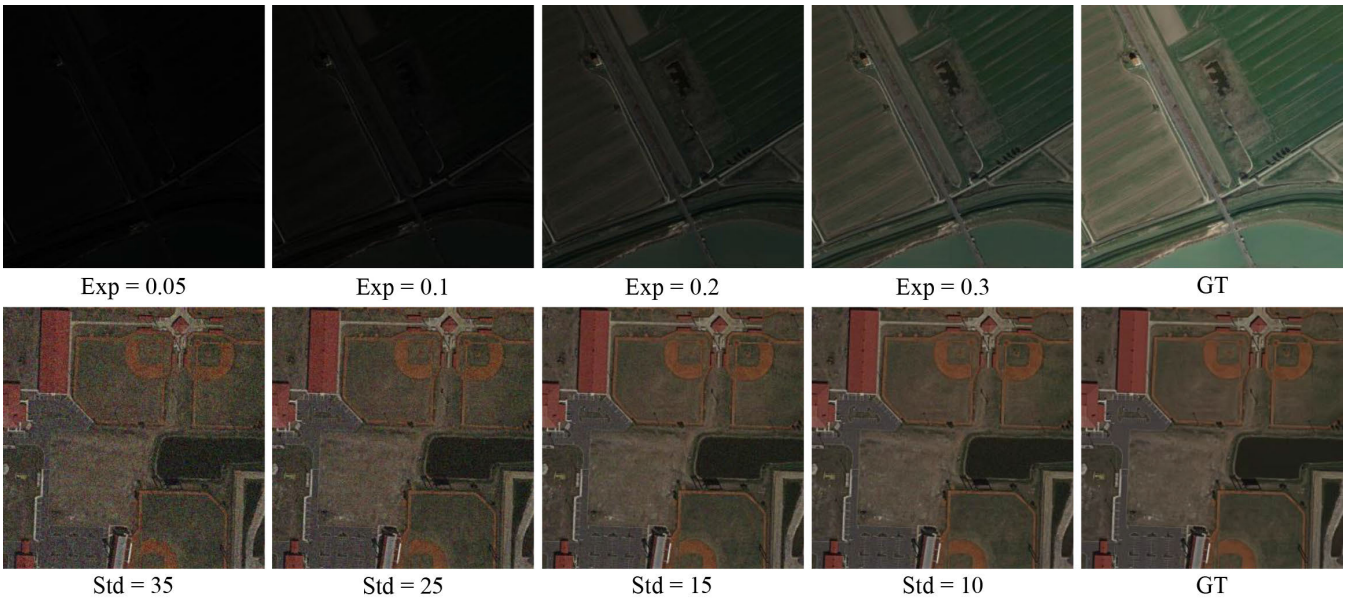


Fig. 7. Proposed iSAID-dark dataset contains low-light images with varying exposure levels and noise levels.

TABLE II

COMPARISON WITH STATE-OF-THE-ART LLIE METHODS ON THE NO-REFERENCE DATASETS, ALONG WITH THE NUMBER OF PARAMETERS, GFLOPS, AND FPS ON IMAGES OF 1080P RESOLUTION. THE BEST RESULTS ARE INDICATED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	Venue	Exdark	DICM	NPE	LIME	darks	Ave.	FPS	FLOPs(G)	Params (M)
Zero-DCE++ [46]	TPAMI'22	3.992	3.924	3.489	4.188	3.760	3.870	281	**0.21**	0.0106
SCI [56]	CVPR'22	3.968	3.657	3.912	4.165	5.450	4.230	**293**	1.1	**0.0004**
SNR [13]	CVPR'22	4.125	4.458	5.503	5.844	3.456	4.677	3	880	40.0842
LLFormer [57]	AAAI'23	3.960	3.890	3.629	4.175	4.502	4.031	-	-	24.5183
CUE [58]	ICCV'23	3.989	3.722	3.977	4.068	3.708	3.892	3	198	0.2407
FourLLIE [28]	ACMM'23	3.941	3.685	3.550	4.056	4.119	3.870	6	515	1.4855
LANet [59]	IJCV'23	4.322	3.992	3.828	4.493	3.478	4.022	15	317	0.0422
NeRCo [60]	CVPR'23	4.092	4.052	4.048	4.007	4.140	4.067	3	1341	7.8414
Ours	-	**3.673**	**3.540**	**3.347**	**4.003**	**3.258**	**3.564**	4	501	2.5765

A. Experiment Details

1) Datasets and Evaluation Metrics: We first validate the performance of the proposed method on the iSAID-dark and darkrs datasets. In addition, to assess the generalization ability of our model to low-light images in other scenarios, we further evaluate the performance of the proposed method on five widely used datasets, including a reference-based dataset (LOL [62]) and three no-reference datasets (DICM, Exdark, and NPE).

We evaluate the visual quality of reference-based datasets using PSNR [16], SSIM [15], and LPIPS [63] metrics. For no-reference images, we assess their naturalness using the NIQE [14] metric. Higher PSNR and SSIM scores indicate better quality, while lower NIQE and LPIPS scores are desirable.

2) Compared Methods: To evaluate the performance of the proposed method, we compare DFFN with current state-of-the-art methods, including a curve-based method (Zero-DCE++ [46]), two unsupervised learning-based methods (SCI [56] and NeRCo [60]), and five supervised learning-based methods (LLFormer [57], SNR [13], CUE [58], FourLLIE [28], and LANet [59]). In the experiments, we retrain and test these methods using the weights and publicly available code provided by these approaches (for the proposed iSAID-dark dataset) for fair comparison.

3) Implementation Details: Our model is trained on the LOL dataset and the iSAID-dark dataset separately. To facilitate better convergence of the model, we set the initial learning rate to 0.0002, and then linearly decrease it by half every 100 epochs, with a total of 500 epochs for training. We use the ADAM optimizer for model optimization. All experiments are conducted on a Windows computer equipped with an NVIDIA RTX 3090 (24 GB RAM) for training and testing. In addition, we set the batch size to 12, randomly crop each image into 256 $\times$ 256 patches, and augment these patches using data augmentation techniques such as horizontal and vertical flips. For image features, we set C to 20, and subsequently, the feature levels are transformed to 40, 80, and 160.

B. Comparison With Stat-of-the-Art Methods

1) Quantitative Results: We first quantitatively compare DFFN with state-of-the-art methods. Table I presents the PSNR, SSIM, and LPIPS results on reference-based datasets. From Table I, it can be seen that our method achieved the

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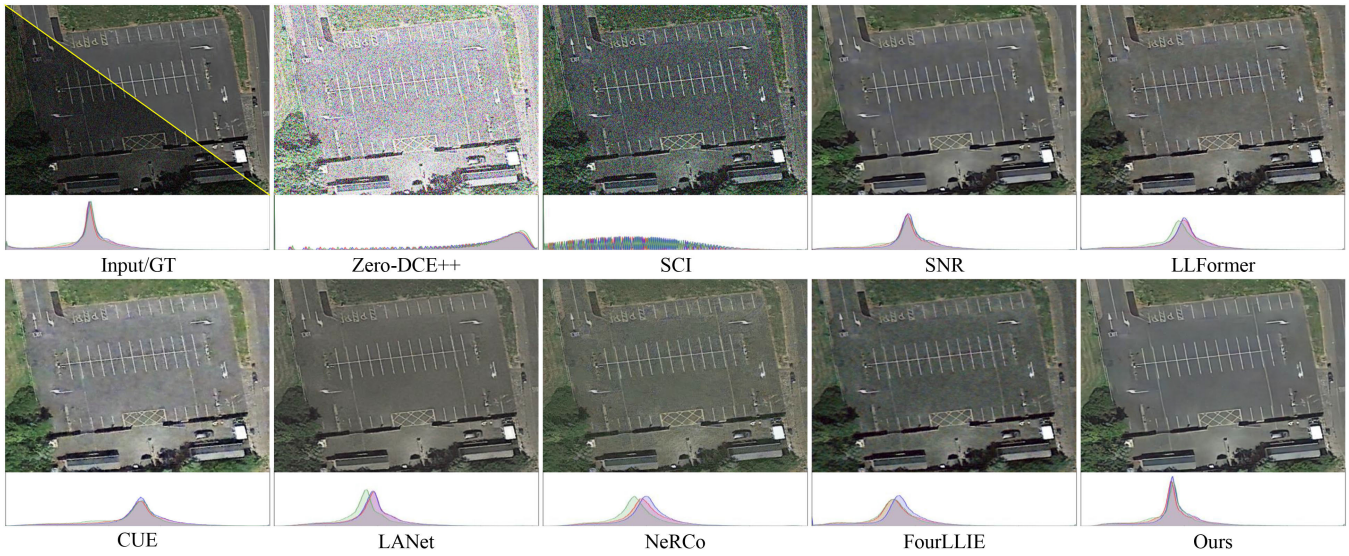


Fig. 8. Visualization results on the iSAID-dark dataset. We present the histogram of color distribution for the images. The histograms placed in input/GT represent the color distribution of the GT. It can be observed that our method's histogram is closer to the GT histogram.

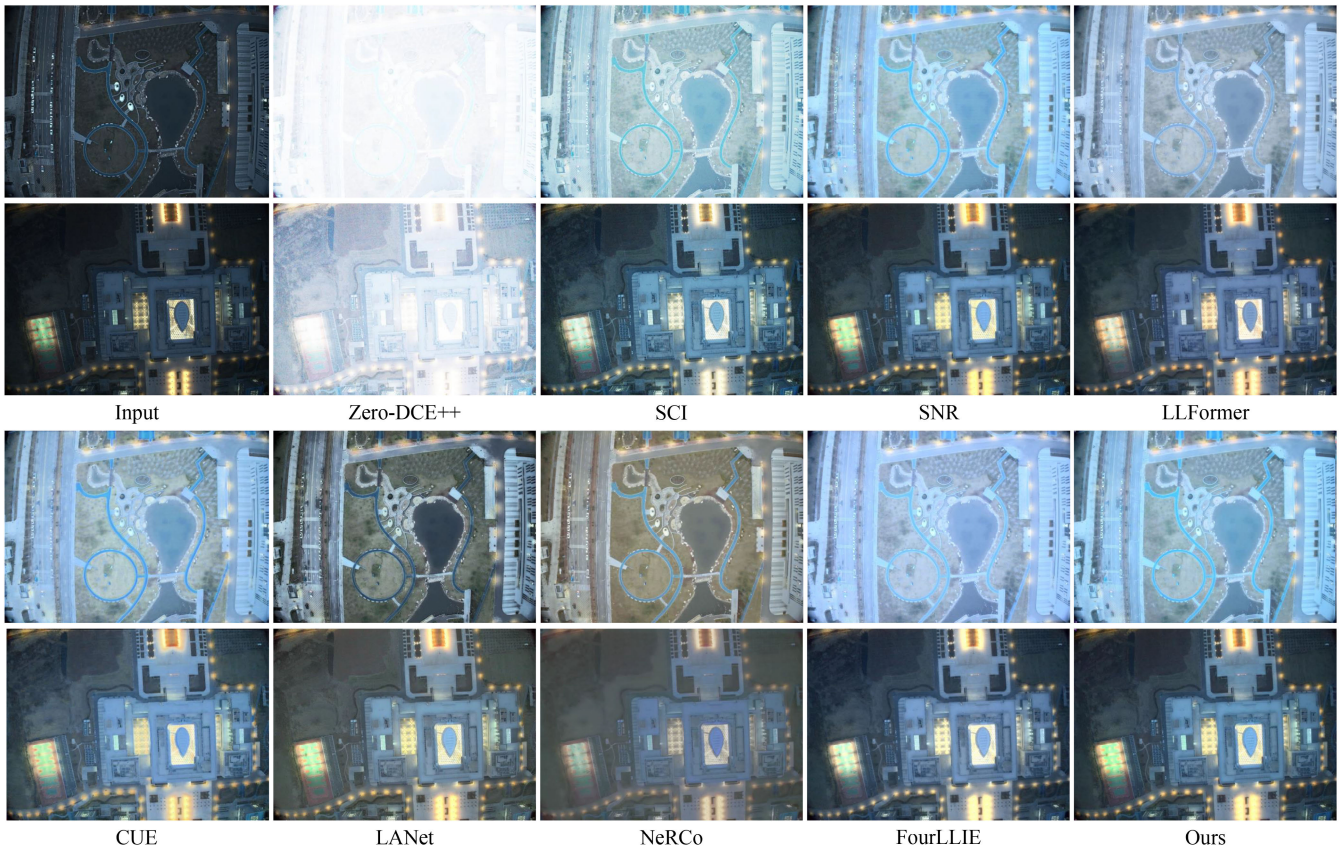


Fig. 9. Visualization results on the darkrs dataset.

best results on both reference datasets. It is worth noting that the results shown for iSAID-dark are obtained by testing with weights trained on LOL, indicating that models trained on existing datasets may not be suitable for low-light RS image enhancement.

Table II presents the NIQE results on the no-reference dataset. It can be observed from Table II that our method achieves the best on all datasets, which fully demonstrates the superiority of DFFN. We also present the parameter count of the models in Table II. The results indicate that although our method cannot compete with no-reference methods in terms of complexity, the performance of no-reference methods on low-light RS datasets is extremely unsatisfactory (for example, SSIM of Zero-DCE++ on iSAID-dark is only 0.055).

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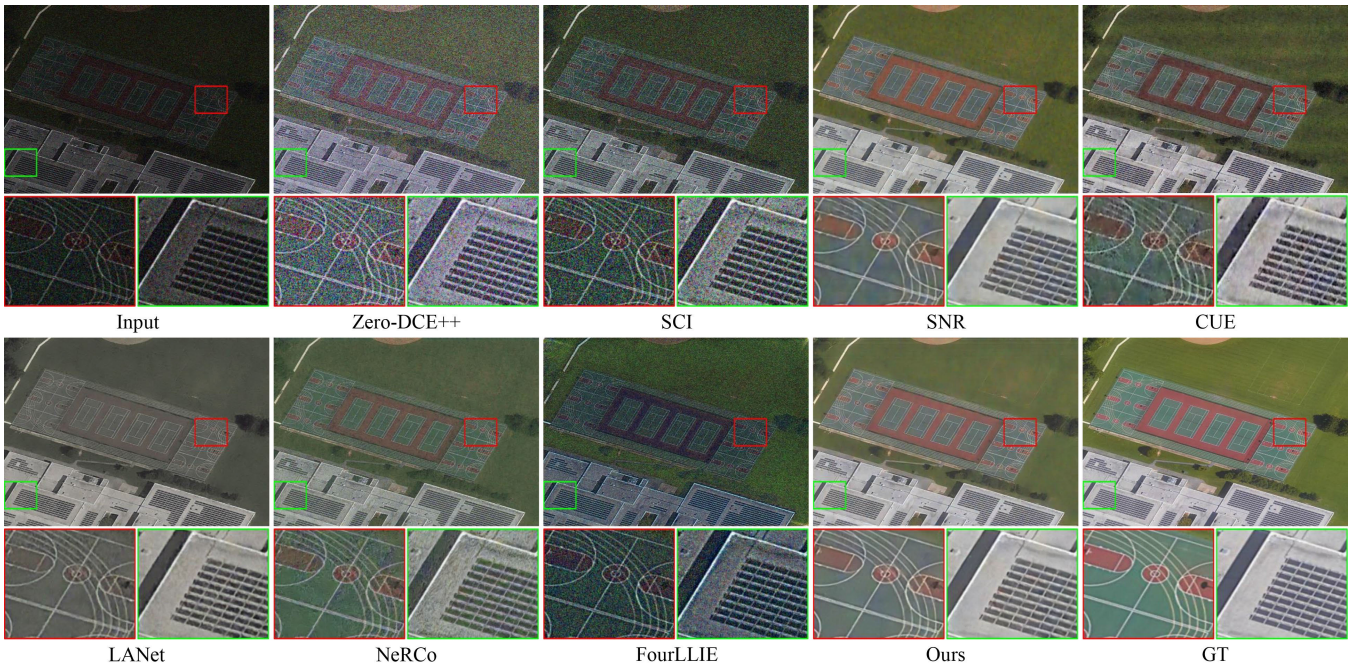


Fig. 10. Visual results on the 1080p image. The selected area is magnified to show details.

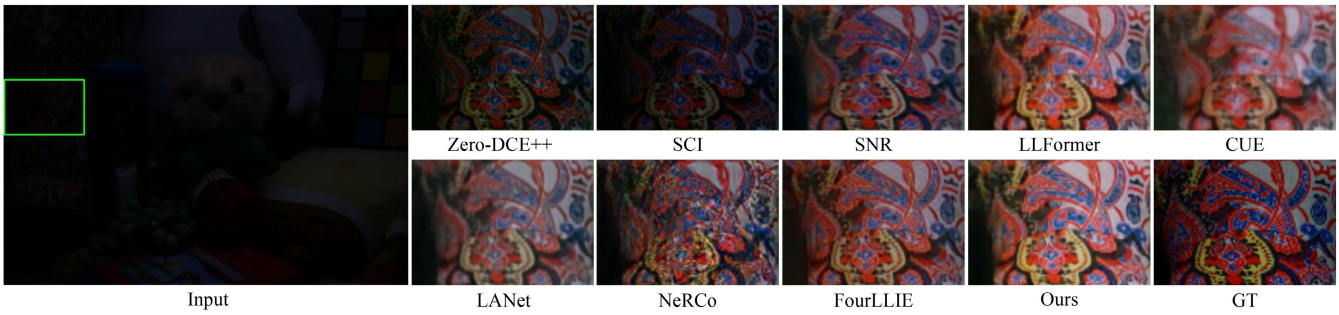


Fig. 11. Visualization results on the LOL dataset. The selected area is magnified to show details.

In contrast, our method achieves a remarkable balance between model complexity and performance.

2) Qualitative Results: Fig. 8 illustrates the enhancement results of our method compared to other state-of-the-art methods on the iSAID-dark dataset. In addition, histograms of color distributions for these images are plotted, where the histograms shown under input/GT represent the GT histograms. Due to the significant Gaussian noise added to the iSAID-dark dataset, no-reference methods such as Zero-DCE++ and SCI, which were not designed to address denoising issues, still exhibit substantial noise in the recovered images. Consequently, these two methods show low SSIM and PSNR values as indicated in Table I. Furthermore, the restoration results of other methods either suffer from excessive color distortion or exhibit blurred details due to noise. In contrast, our method achieves the best visual results, while the histograms of color distributions are closer to the GT.

Fig. 9 depicts RS images captured by drones in real nighttime environments. The dataset contains numerous areas with uneven exposure, providing a better balance for evaluating the model's ability to enhance low-light RS images in real-world scenarios. It can be observed that Zero-DCE++ exhibits overexposure, while LANet and NeRCo show limited brightening effects, resulting in overall low contrast in the images. SCI, LLFormer, and FourLLIE demonstrate good enhancement in low-light areas, but some regions suffer from color distortion and FourLLIE blurs image details. Our method effectively enhances image brightness while better-preserving image color.

As shown in Fig. 10, for high-resolution images, Zero-DCE++, SCI, and FourLLIE struggle to restore the image brightness and fail to address noise issues. CUE, LANet, SNR, and NeRCo exhibit unexplained artifacts or neglect color information. LLFormer, due to its large computational load, cannot process images of this resolution on our equipment. In contrast, our method maintains outstanding performance on high-resolution images.

Fig. 11 displays the enhancement results on the LOL dataset. We have selected certain regions within the images for zoomed-in view to present the image details more clearly. From the image, it can be seen that Zero-DCE++, SCI, and SNR only achieved limited enhancement. The results of CUE

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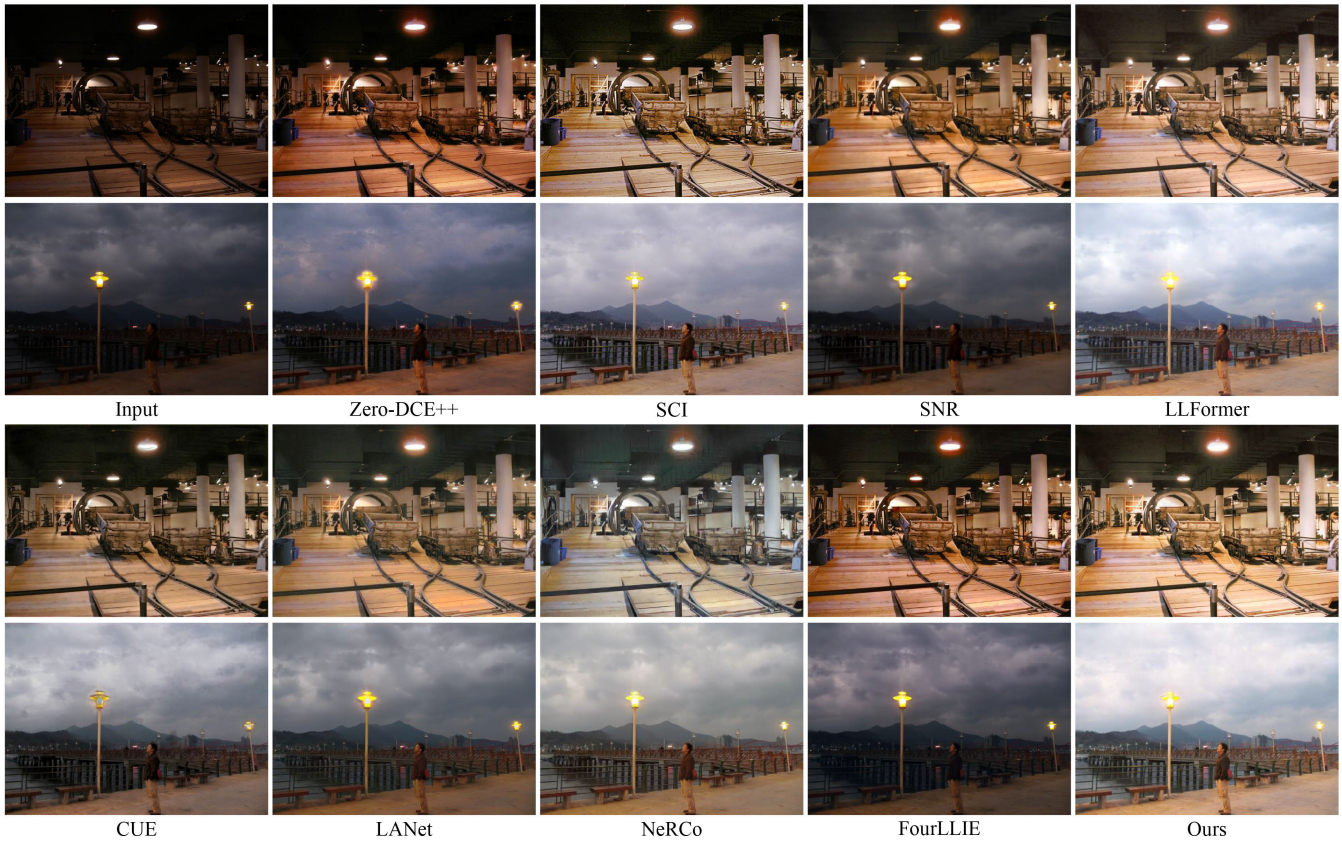


Fig. 12. Visualization results on the DICM dataset (top) and the NPE dataset (bottom).

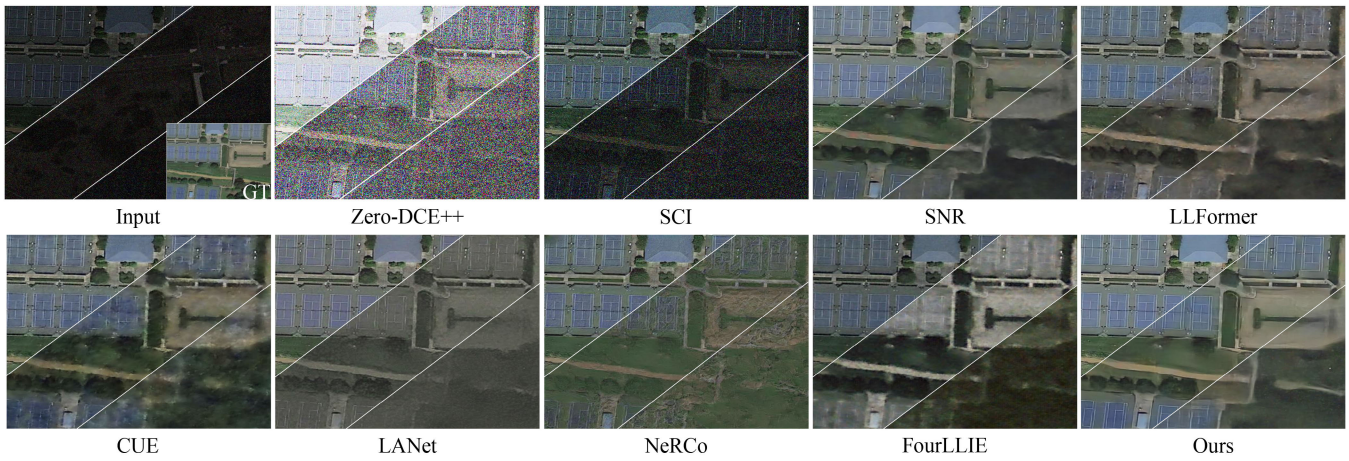


Fig. 13. Qualitative enhancement results of three low-light images of the same scene. (Left to right) Brightness of the images gradually decreases, and the noise concentration gradually increases.

and LANet are too blurred in detail. NeRCo exhibited some unexplained shadows. Although FourLLIE and LLFormer also achieved decent enhancement effects, compared to them, our method shows more prior colors and clearer details.

Fig. 12 illustrates the enhancement results on the DICM dataset and the NPE dataset. Zero-DCE++, SNR, and FourLLIE produce overly dark enhancement results, with little difference compared to the degraded images, especially in the images of the second row. SCI, CUE, LANet, and NeRCo introduce some color deviations and inadequate brightness restoration in the enhanced images. In comparison, our method effectively suppresses noise while restoring image brightness, presenting the most natural images.

Meanwhile, we tested the recovery ability of the method in different lighting and noise environments, as shown in Fig. 13. Zero-DCE++ and SCI exhibit significantly different recovery capabilities under varying illumination conditions, and their noise repair ability is poor. LANet and FourLLIE struggle to capture color information in extremely dark environments. Due to excessive noise interference, the detail recovery capabilities

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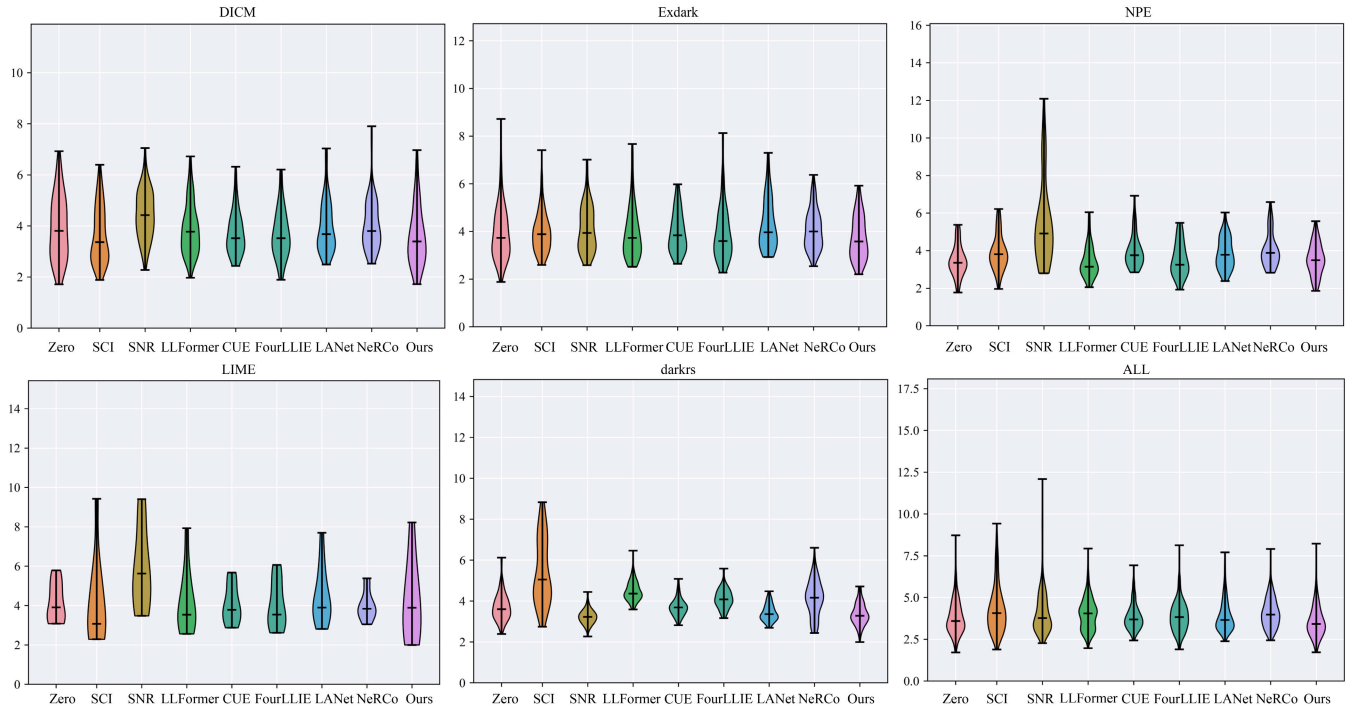


Fig. 14. On no-reference image datasets, the numerical distribution of NIQE test results varies across different enhancement methods. The evaluation value of the proposed method is relatively concentrated, with low scores across multiple images, and better performance in low-light enhancement scenarios.

of CUE and NeRCo are greatly affected. Although LLFormer and SNR, based on the Transformer architecture, demonstrate relatively excellent hierarchical correction capabilities, their performance is still unsatisfactory under excessively dark and noisy conditions. In contrast, our method achieves excellent recovery results in all three levels of adverse environments.

Fig. 14 displays the NIQE distributions of the comparison methods across five test datasets. It is evident that DFFN exhibits a more concentrated data distribution, which strongly indicates the superior generalization capability of our model across various scenarios. In addition, when all test set data is combined and plotted together (denoted as ALL), DFFN's data appears more centralized. This further demonstrates the exceptional stability of our model.

C. Ablation Study

In this section, we conducted extensive ablation experiments to validate the effectiveness of the proposed modules, the two-stage network, and the fusion of spatial and frequency domains. Subsequently, we evaluated the performance of networks with different depths and the use of multilevel loss. Finally, we replaced the frequency domain block with the state-of-the-art transformer block to validate the effectiveness of the proposed model.

1) Contribution of Two-Stage Architecture: In this section, we validate the effectiveness of the proposed two-stage network architecture. All experiments are conducted based on the foundational model without employing the IFAM modules. M_a represents the use of spatial information only in a single stage, M_b denotes the simultaneous utilization of spatial and frequency domain information in the single stage, and M_c employs the two-stage architecture, but with O_1 as the input for the second stage, M_d is identical to our training architecture.

As shown in Table III, the performance of M_a is the poorest, indicating that solely recovering spatial information in a single stage may yield unsatisfactory results. Compared to M_b, M_c, and M_d exhibit significantly improved performance, implying that decoupling and separately learning degraded information is crucial for information recovery. Moreover, M_d achieves a gain of 0.21 dB in PSNR over M_c, indicating that preserving phase consistency is more critical for learning in the second stage.

2) Contribution of Dual-Domain Fusion: In this section, we discuss the effectiveness of integrating spatial and frequency domain information and the effectiveness of different integration methods, while also exploring the effectiveness of the proposed IFAM and DDAB/DDPB. M_1 denotes the use of spatial information only in a two-level network. M_2 and M_3 adopt a sequential method to learn spatial and frequency domain information, respectively. Specifically, M_2 first learns spatial information (as shown in Fig. 15), while M_3 first learns frequency domain information. M_4 has the same training architecture as ours, but none of the four configurations use IFAM. M_5 indicates the absence of the proposed DDAB/DDPB.

As shown in Table IV, the model M_1 utilizing only spatial information performs poorly, indicating the effectiveness of integrating spatial–frequency dual-domain information. Compared to M_2, M_3, and M_4, which learns in parallel, achieves the best performance in most metrics. This suggests that compared to the parallel approach, serial learning is more prone to losing information from one domain while learning information from

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TABLE III

ABLATION STUDY OF THE TWO-STAGE ARCHITECTURE DESIGN INCLUDES EXPERIMENTS WHERE "FREQUENCY" INDICATES WHETHER FREQUENCY DOMAIN IS USED, "STAGE2" DENOTES WHETHER THE SECOND STAGE IS UTILIZED, AND "SWAP" INDICATES WHETHER THE PHASE INFORMATION OF THE OUTPUT O_1 FROM THE FIRST STAGE IS REPLACED

Model	Frequency	Stage2	Swap	PSNR↑	SSIM↑	LPIPS↓
M_a	✗	✗	✗	21.58	0.734	0.198
M_b	✓	✗	✗	22.59	0.750	0.178
M_c	✓	✓	✗	24.52	0.771	0.167
M_d	✓	✓	✓	**24.93**	**0.772**	**0.166**

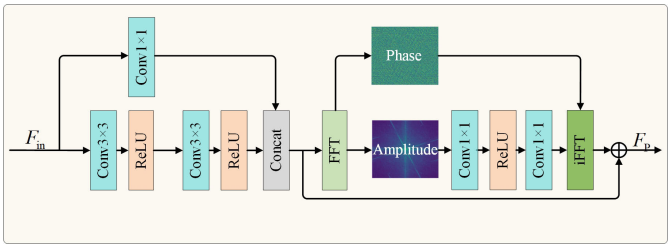


Fig. 15. Modified DDAB first learns spatial domain information in a serial manner, followed by learning frequency domain information.

TABLE IV

ABLATION STUDY ABOUT THE COMBINATION OF SPATIAL–FREQUENCY DOMAIN FEATURES

Model	M_1	M_2	M_3	M_4	M_5	Ours
PSNR↑	21.55	24.88	24.88	24.93	22.42	**25.38**
SSIM↑	0.655	0.771	0.772	0.772	0.701	**0.773**
LPIPS↓	0.289	**0.162**	0.171	0.166	0.237	0.165

another. By employing parallel computation, this information loss is effectively mitigated. When DDAB/DDPB is not employed, the model's performance declines significantly, highlighting the superiority of the proposed DDAB/DDPB.

Moreover, when IFAM is utilized, there is a significant improvement in PSNR, indicating that IFAM successfully aggregates features from different sources and enhances the global context representation of the network.

3) Contribution of Different Depths: In this section, we investigate the impact of network depth on performance. Specifically, we set the number of channels expanded after the first convolutional layer to C and experiment with $C = 10$, 20, and 30. For these three configurations, we train the network for 1000 epochs. As shown in Fig. 16, the left graph illustrates the loss decline curve on the validation set, while the right graph shows the PSNR improvement curve.

When $C = 10$, although the network has fewer parameters and faster inference speed, it is too shallow, resulting in slow fitting speed and poor learning quality. When $C = 30$, the network is too deep with more parameters, leading to overfitting and poor performance. In addition, the network has an excessive number of parameters when $C = 30$, as shown in Table V. Comparatively, the network achieves optimal performance when $C = 20$.

TABLE V

ABLATION STUDY FOR NETWORKS WITH DIFFERENT DEPTHS. $C = 10$ MEANS THE INITIAL FEATURE CHANNELS ARE 10, WHICH THEN EXPAND SEQUENTIALLY TO 2C, 4C, AND 8C WITHIN THE NETWORK

Depth	PSNR↑	SSIM↑	LPIPS↓	FPS	FLOPs(G)	Params(M)
C = 10	21.58	0.734	0.222	**7**	**128**	**0.6412**
C = 20	**25.30**	**0.784**	**0.151**	4	501	2.5765
C = 30	24.90	0.735	0.176	2	1121	5.7883

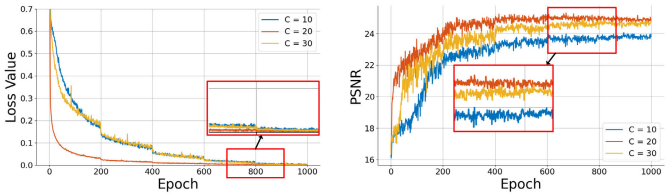


Fig. 16. Ablation study on different depths. C represents the initial number of channels expanded after entering the network, and it can be seen from the figure that the network performs best when $C = 20$.

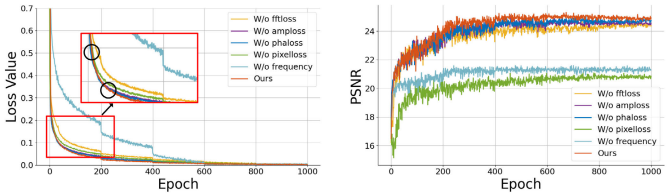


Fig. 17. Ablation study on multistage loss. (Left) Loss decrease curves, and (right) PSNR increase curves. Each experiment was conducted for 1000 epochs.

4) Contribution of Multistage Loss: In this section, we validated the benefits of multilevel loss for the proposed model. As shown in Fig. 17, the left graph depicts the loss decline curves when using different losses, and the right graph illustrates the PSNR increase curves. "W/o frequency" indicates the absence of all frequency domain losses, while "W/o pixelloss" represents the absence of spatial domain loss. The remaining configurations demonstrate the results when each of the three frequency domain losses is individually omitted. From the enlarged area of the left graph, it can be observed that when only spatial or frequency domain information is used for supervision, the fitting speed is significantly slower. When both spatial and frequency domain information are used for supervision simultaneously, the convergence speed of the model is similar initially, but after 100 epochs, the model with the complete loss converges significantly faster. Notably, in the case of "W/o fftloss," although the model uses the corresponding frequency domain information for supervision in both stages, the lack of overall frequency domain guidance makes it more difficult for the model to fit, and the final performance is worse.

5) Compared With Transformer Blocks: Finally, we evaluated the effectiveness and efficiency of the proposed DFFN compared to transformers. Specifically, we replaced the frequency domain branches in DDAB and DDPB with the most advanced transformer blocks [64] currently available, and only used spatial domain information for supervision, with all other operations being the same as in DFFN.

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TABLE VI

ABLATION STUDY FOR USING TRANSFORMER BLOCKS, STB INDICATES THAT EACH BLOCK USES ONLY ONE SELF-ATTENTION BLOCK AND ONE FFN, WHILE MTB SIGNIFIES THAT MULTIPLE SELF-ATTENTION BLOCKS AND FFNS ARE STACKED WITHIN EACH TRANSFORMER BLOCK

Model	Attention	Head	PSNR↑	SSIM↑	LPIPS↓	FPS	FLOPs(G)	Params (M)
STB	[1,1,1,1]	[2,2,2,2]	23.53	0.687	0.260	1	1081	6.3139
MTB	[2,4,6,8]	[2,2,4,4]	23.02	0.726	0.203	-	-	26.9895
Ours	-	-	**25.38**	**0.788**	**0.146**	**4**	**501**	**2.5765**

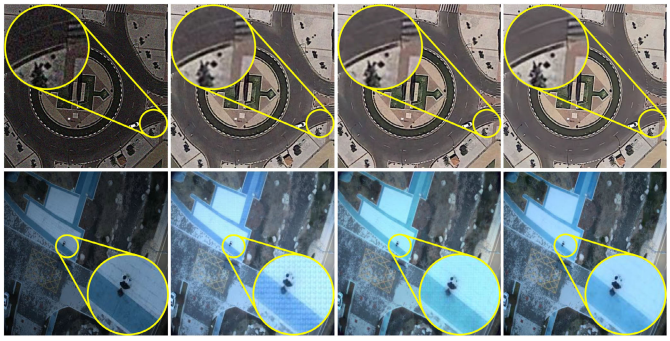


Fig. 18. Visualization results compared with transformer blocks, STB indicates that each block uses only one self-attention block and one FFN, while MTB signifies that multiple self-attention blocks and FFNs are stacked within each transformer block.

Initially, to ensure fairness, we used a single self-attention block and a single FFN operation in each block, setting num_head to 2. We refer to this as the single transformer block (STB). Although transformers are designed to capture long-term dependencies, shallower transformer blocks may struggle to learn rich feature representations, resulting in inferior performance compared to frequency domain networks when processing large images. As shown in Table VI, using only a single operation in transformer blocks leads to suboptimal results while introducing a significant number of parameters.

Subsequently, we followed the mainstream settings, stacking self-attention blocks and FFN operations in each stage as [2,4,6,8], with num_head set to [2,2,4,4], which we refer to as multiple transformer block (MTB). However, this setup results in nearly 27M parameters, and the final enhancement effect is still slightly inferior to our method. In addition, we measured the FLOPs and FPS on 1080p images for STB and our method (MTB could not run due to excessive computational requirements). In comparison, our frequency domain strategy has only 2.57M parameters, with 501 GFLOPs and 4 FPS, achieving the best results. In addition, as shown in Fig. 18, the visualization demonstrates that our method achieves the best visual effects.

VI. CONCLUSION

In this article, we propose a two-stage DFFN for enhancing low-light RS images. The DFFN combines the advantages of spatial and frequency domains, consisting of two subnetworks: the amplitude illumination stage and the phase refinement stage. To better integrate dual-domain information, we design the corresponding DDAB & DDRB. Due to the lack of information interaction in the two-stage network, we further design IFAM to interactively fuse features of different domains, stages, and sizes in the two stages, enhancing the model's contextual representation capability. Furthermore, we introduce the iSAID-dark and darkrs datasets for low-light RS image enhancement.

Although our method demonstrates good performance in most cases, it still has certain limitations. The presence of numerous FFT and iFFT processes in the network lowers the model's inference speed. In addition, when there is a significant amount of noise in the environment, DFFN struggles to remove all the noise, which is a common challenge faced by current mainstream LLIE methods. In the future, we will focus on addressing the denoising problem in extremely low-light environments while improving the model's inference speed, so that our method can be better applied to real-world scenarios.

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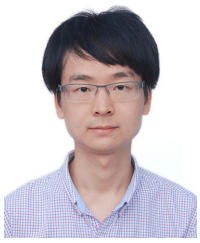
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